

Performance review of nonlinear shallow water equations in tsunami wave modeling

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Summary

This research examines and evaluates the nonlinear shallow water equations in the numerical modeling of tsunami waves. Tsunami modeling comprises three primary phases: the generation of tsunamis, their propagation across sea, and the final run-up upon reaching coastal regions. To evaluate the efficacy of the nonlinear shallow water model, an initial benchmark test was conducted. This test was pivotal in establishing the model's reliability, which then paved the way for simulating three significant historical tsunami events—the devastating 2004 Indian Ocean tsunami, the catastrophic 2011 tsunami in Japan, and the 1945 Makran tsunami. Laboratory data plays a crucial role in tuning and calibrating the numerical models, enabling them to better reflect real-world behaviors and outcomes. The results from these simulations showed a noteworthy alignment with actual historical data, showcasing an acceptable level of accuracy in the nonlinear shallow water numerical modeling approach. This compatibility was particularly pronounced for the 2011 Japan tsunami, where the availability of more contemporary and precise data supported more accurate modeling results. However, it is crucial to note that the observed discrepancies between the model outputs and the real-world scenarios cannot be entirely attributed to instrumental errors. Instead, a significant factor contributing to these discrepancies lies in the incomplete understanding of the underlying sources and mechanics of tsunami generation. Given this complexity, modeling tsunami generation carries a high level of uncertainty and represents a critical aspect of the modeling process. While the foundational equations governing water waves are well-established, there remains a pressing need for advancements in the modeling techniques utilized for tsunami propagation and run-up. Specifically, there is a strong imperative to refine the accuracy with which factors such as spray dynamics, friction with the seabed and coastal environments, and the multifaceted behavior of waves upon making landfall are incorporated into these models. These advancements have important effects beyond just research. They significantly impact tsunami risk assessment, early warning systems, and preparedness in coastal communities. Effective tsunami modeling not only enhances our understanding of these devastating natural phenomena but also fortifies our readiness to respond to future events. Given that tsunami modeling is a prerequisite for studying tsunami risk and developing warning systems in tsunami-prone areas, one essential requirement for accurate modeling is high-resolution bathymetric and topographic data. Furthermore, to better understand tsunamis, future studies must aim for a clearer picture of the geometry of tsunami sources, leading to more precise estimates of seabed deformation alongside conducting thorough geophysical, geological, remote sensing, and other field studies. The experiences from the 2004 Indian Ocean tsunami and the 2011 Japan tsunami demonstrate that utilizing tsunami early warning technology is crucial for providing timely alerts before the waves reach the shore, as there is no time to escape once the waves have arrived. In the two mentioned tsunamis, had a warning system been in place or functioning correctly, there would have been an appropriate opportunity to issue tsunami alerts from the source area to surrounding coastlines.

Keywords: Tsunami, numerical modeling, shallow water equations, benchmark test, tsunami warning

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